

PALLOR AND ANAEMIA

- Patient has pallor if s/he has pale conjunctiva or palms. Compare patient's palms to your own.
- Check Hb: **anaemia** likely if:
 - Non pregnant woman has Hb < 12.
 - Pregnant woman has Hb < 11 → 160.
 - Man has Hb < 13.

Give urgent attention to the patient with pallor/anaemia and any of:

- Hb < 6
- BP < 90/60
- Swollen legs
- Widespread/easy bruising
- Pulse ≥ 100
- Dizzy/faint
- Jaundice
- Purple/red rash that does not disappear with pressure
- Respiratory rate ≥ 30
- Chest pain or palpitations
- Black¹ or bloody stools

Manage and refer urgently:

- If respiratory rate increased, give face mask oxygen.
- If BP < 90/60, give **sodium chloride 0.9%** 500mL IV over 30 minutes, repeat until systolic BP > 90. Continue 1L 6 hourly. Stop if breathing worsens.

Approach to the patient with pallor/anaemia not needing urgent attention

- Test for HIV ↗ 110 and TB ↗ 92.
- Exclude pregnancy ↗ 157.
- If fever now or in past 3 days, and in a malaria area in past 3 months, arrange same day malaria test². If positive, **malaria** likely → 24.
- If not pregnant, send full blood count (FBC) and manage further according to mean cell volume (MCV)³ result:

MCV³ low

Iron deficiency anaemia likely

Is patient a man or a woman who no longer has periods?

Yes

Discuss/
refer to
look for
hidden
blood loss.

No

- Ask about abnormal vaginal bleeding: if abnormal ↗ 57.
- Give **ferrous sulphate compound BPC** 170mg or **ferrous fumarate** 200mg 12 hourly with food. If not tolerated (abdominal pain, nausea, vomiting, constipation), give instead **ferrous sulphate compound BPC** 340mg or **ferrous fumarate** 400mg once weekly with food
 - Repeat Hb monthly on treatment: if Hb decreases or if no better after 4 weeks, refer.
 - Continue treatment until 3 months after Hb reaches normal value.
- Advise:
 - To eat foods rich in iron: liver, kidney, meat, eggs, spinach, beans, peas, lentils, nuts, dried fruit and fortified cereals. Foods rich in vitamin C help iron absorption: guavas, peppers, oranges, strawberries, broccoli, cauliflower.
 - Avoid drinking tea/coffee with meals as these interfere with iron absorption. Also avoid taking iron tablets with milk or calcium tablets.
 - Warn that stools may become black with treatment, reassure this is normal.

MCV³ normal

Systemic disease or chronic condition likely

- If HIV, TB and pregnancy excluded, discuss/refer.
- If patient is known with life-limiting illness, also consider giving palliative care ↗ 170.

MCV³ high

Macrocytic anaemia likely

Patient postpartum or known to misuse alcohol⁴?

Yes

Folate deficiency likely

- Review medication: if on zidovudine or anticonvulsants, discuss with doctor.
- Give **folate** 5mg daily until Hb normal.
- Repeat Hb monthly on treatment: if Hb decreases or if no better after 4 weeks, refer.
- Advise:
 - To eat foods rich in folic acid: liver, eggs, fortified cereals, citrus fruit, spinach, other green vegetables, lentils, dry beans, peanuts.
 - Avoid alcohol ↗ 142.
- If chronic diarrhoea, refer.

No

Refer to
investigate for
vitamin B12
deficiency.

¹Black stools may be caused by iron tablets. Only refer if black stools started before iron treatment. ²Test for malaria with rapid diagnostic test if available, and parasite slide microscopy. ³Mean cell volume (MCV) helps identify cause of anaemia. Check on FBC result sheet if MCV low, normal or high compared to reference range. ⁴Drinks > 14 drinks/week or ≥ 4 drinks/session. One drink is 1 tot of spirits, or 1 small glass (125mL) of wine or 1 can/bottle (330mL) of beer.